

PHILIP EUGEN NESBAKKEN

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EDUCATION

Boston University

Sep 2025 - May 2026

Master of Arts, Mathematics

Boston, MA

- Thesis: *Brownian Motion on Lie Groups*; studies how stochastic processes behave on smooth non-Euclidean spaces using Lie group structure, heat kernel analysis, and representation theory
- Coursework: Graduate Probability, Graduate Abstract Algebra, Mathematical Foundations of Neural Networks, Graduate Complex Analysis

Boston University

Sep 2022 - May 2025

Bachelor of Arts, Computer Science & Mathematics

Boston, MA

- GPA: 3.63; graduated one year early; Dean's List
- Graded weekly problem sets and exams for 110+ undergraduate students in Multivariable Calculus and Discrete Mathematics (Jan - May 2025)
- Coursework: Probability Theory, Algorithms & Data Structures, Computational Statistics, Linear Algebra, Real Analysis, Discrete Mathematics, Operating Systems, Computer Organization

PROFESSIONAL EXPERIENCE

Endor Global

Aug 2025 - Jun 2026

Clojure Fullstack Developer/Data Science

Remote

- Built features across the platform in Clojure and ClojureScript on a 5-engineer team, owning work end to end from design through weekly production deployments
- Jointly maintained the app's entire data pipeline and wrote analytical tooling in a functional style, using immutable data structures and pure functions throughout
- Helped maintain CI/CD pipelines supporting weekly releases for the remote-first team
- Wrote notebooks with actionable data points in Clojure and SQL; product-first view

Subsea7

Jun 2025 - Aug 2025

Market Intelligence Intern

Houston, TX

- Built Power BI dashboards and Excel models to project market share and revenue across Oil & Gas and Renewables; pulled data from multiple industry sources and reconciled inconsistencies
- Presented competitive intelligence findings to senior leadership; the analysis fed directly into long-term strategic planning discussions
- Researched offshore energy trends using industry reports, trade publications, and public filings

Dintero

Jun 2024 - Aug 2024

Backend Developer Intern

Oslo, Norway

- Developed and maintained backend services in TypeScript for a live Norwegian payment platform processing double-digit-billion-NOK annual transaction volume
- Designed and shipped 5 REST API endpoints with full test coverage, working closely with a 5-engineer team spanning frontend and product
- Fixed 30+ production bugs affecting payment reliability and wrote regression tests to keep them fixed
- Worked within a real engineering workflow: code review, staging deployments, on-call exposure

Boston University

Jan 2025 - May 2025

Student Grader, Mathematics

Boston, MA

- Graded problem sets and exams for 110+ undergraduate students across Multivariable Calculus and Discrete Mathematics
- Wrote feedback on student work and flagged recurring misconceptions to the course instructor

University of Oslo

Data Analyst Intern

Apr 2020 - May 2022

Oslo, Norway

- Analyzed clinical trial data from the EPoS colorectal cancer studies (15,581 patients across 38 endoscopy centres in 5 European countries) using Stata; helped design the statistical methodology and clean multi-site datasets
- Sat in on weekly methodology discussions with senior researchers and biostatisticians over two years
- Co-authored a peer-reviewed publication on colonoscopy frequency outcomes from the EPoS trials (2022), since cited by 6 subsequent papers

PROJECTS

Geometric Brownian Motion Simulator | github.com/eugnes03/GBM | C++, Python

- C++ library that simulates asset price paths via Geometric Brownian Motion; supports Black-Scholes option pricing (analytical and Monte Carlo), path-dependent payoffs, and configurable drift and volatility
- Python layer handles visualization, parameter sweeps, and convergence checks against closed-form solutions

MathAtlas | eugnes03.github.io/MathAtlas

- Structured knowledge base covering 108 mathematical objects across topology and algebraic topology (fundamental groups, homotopy theory, covering spaces, CW complexes)
- Built with KaTeX for typeset math, Mermaid.js for concept dependency graphs, and a fully static architecture with no runtime dependencies
- Designed the schema so theorem dependencies and proof chains are navigable across the whole graph

Sparse Bayesian Neural Networks via Variational Inference

- Implemented a variational Bayesian framework for sparse deep neural networks using spike-and-slab priors, reproducing results from recent NeurIPS research
- Developed Gumbel-softmax relaxations and straight-through estimators to enable efficient optimization of discrete inclusion variables
- Trained and evaluated models on synthetic regression and MNIST, demonstrating probabilistic feature selection and analyzing sparsity-accuracy trade-offs
- Improved training stability through KL annealing, adaptive learning rates, and posterior diagnostics, verifying convergence via ELBO tracking

Personal Webpage & Blog | eugnes03.github.io/personalwebpage

- Multi-stage build: Clay → Quarto → static HTML, deployed via GitHub Pages; fully automated from Clojure source to published site
- Wrote namespaces for theorem/proof rendering, TikZ diagrams, and citation management; posts cover group theory and probability
- Added light/dark theming, dynamic post filtering, and auto-generated indexes via Node.js

PUBLICATIONS

Juul, F., et al. (incl. Nesbakken, P.E.). *Rates of repeated colonoscopies to clean the colon from low-risk and high-risk adenomas: Results from the EPoS trials*. University of Oslo, 2022.

SKILLS

Mathematics: Probability & Stochastic Calculus, Representation Theory, Lie Groups & Differential Geometry, Abstract Algebra, Real Analysis, Statistics

Programming: Python, Clojure/ClojureScript, TypeScript, C/C++, SQL, Bash, Stata

ML & Data: PyTorch, Neural Networks, Variational Inference, Statistical Modeling, Power BI, Excel

Tools: Git, L^AT_EX/TikZ, Quarto, Clay, Linux, REST APIs, CI/CD

Languages: Norwegian (native), English (professional fluency)

Interests: Piano, Waterskiing, Tennis, Mathematical Logic, Commutative Algebra